

Three Perceptual Tools for Seeing and Understanding Visualized Data

Steven L. Franconeri 

Department of Psychology, Northwestern University

Abstract

The visual system evolved and develops to process the scenes, faces, and objects of the natural world, but people adapt this powerful system to process data within an artificial world of visualizations. To extract patterns in data from these artificial displays, viewers appear to use at least three perceptual tools, including a tool that extracts global statistics, one that extracts shapes within the data, and one that produces sentence-like comparisons. A better understanding of the power, limits, and deployment of these tools would lead to better guidelines for designing effective data displays.

Keywords

data visualization, graphic perception, visual communication, graph comprehension, ensemble perception

The table at the top of Figure 1 shows four types of data for six countries in 2015: gross domestic product (GDP) per capita, life expectancy, population, and carbon dioxide (CO₂) emissions per capita. Searching for patterns in those data likely feels sluggish to you because you can read only a handful of numbers per second. And as you compare one handful of numbers with another, your limited working memory capacity causes the original numbers to slip away, or to be overwritten by the new values. Now inspect the data for those same countries in the visualization at the center of Figure 1, which depicts all of those data, plus additional data for GDP and life expectancy for the year 1985 (via the “comet” lines). You should feel the processing power of the visual system that occupies around 40% of your brain (Van Essen et al., 1992), which allows you to rapidly extract statistics across those countries (e.g., China and India have the highest populations; average life expectancy across all countries is around 74 years), to note the roughly linear shape of the arrangement of the bubbles (i.e., GDP is positively correlated with life expectancy), and to efficiently browse specific comparisons (e.g., Bangladesh shows a slightly faster rise in life expectancy compared with India).

Data visualizations are powerful tools for thinking about and communicating with data. Understanding and creating these visualizations are now critical skills for an educated public, even arguably similar in importance

to reading and writing (Börner et al., 2019). To harness that power, a viewer must understand how a set of visual marks, such as bars, lines, or bubbles, can depict data, and how patterns among those marks reflect patterns in the data. In this review, I argue that viewers extract patterns and relationships from visualized data using at least three core perceptual tools—extraction of statistics, extraction of shapes, and comparisons—and demonstrate how the choice of perceptual tool can influence which patterns are seen and how they are understood.

Perceptual Tool 1: Visual Statistics

Data visualizations map numeric data to a set of visual channels, including position, length, area, numerosity, angle, and intensity (for review, see Munzner, 2014). Figure 1 highlights how a single visualization can recruit every one of these channels to depict several types of data simultaneously. Each of these channels has a natural mapping between visual appearance and the data values that are reflected; for example, higher

Corresponding Author:

Steven L. Franconeri, Department of Psychology, Northwestern University
Email: franconeri@northwestern.edu

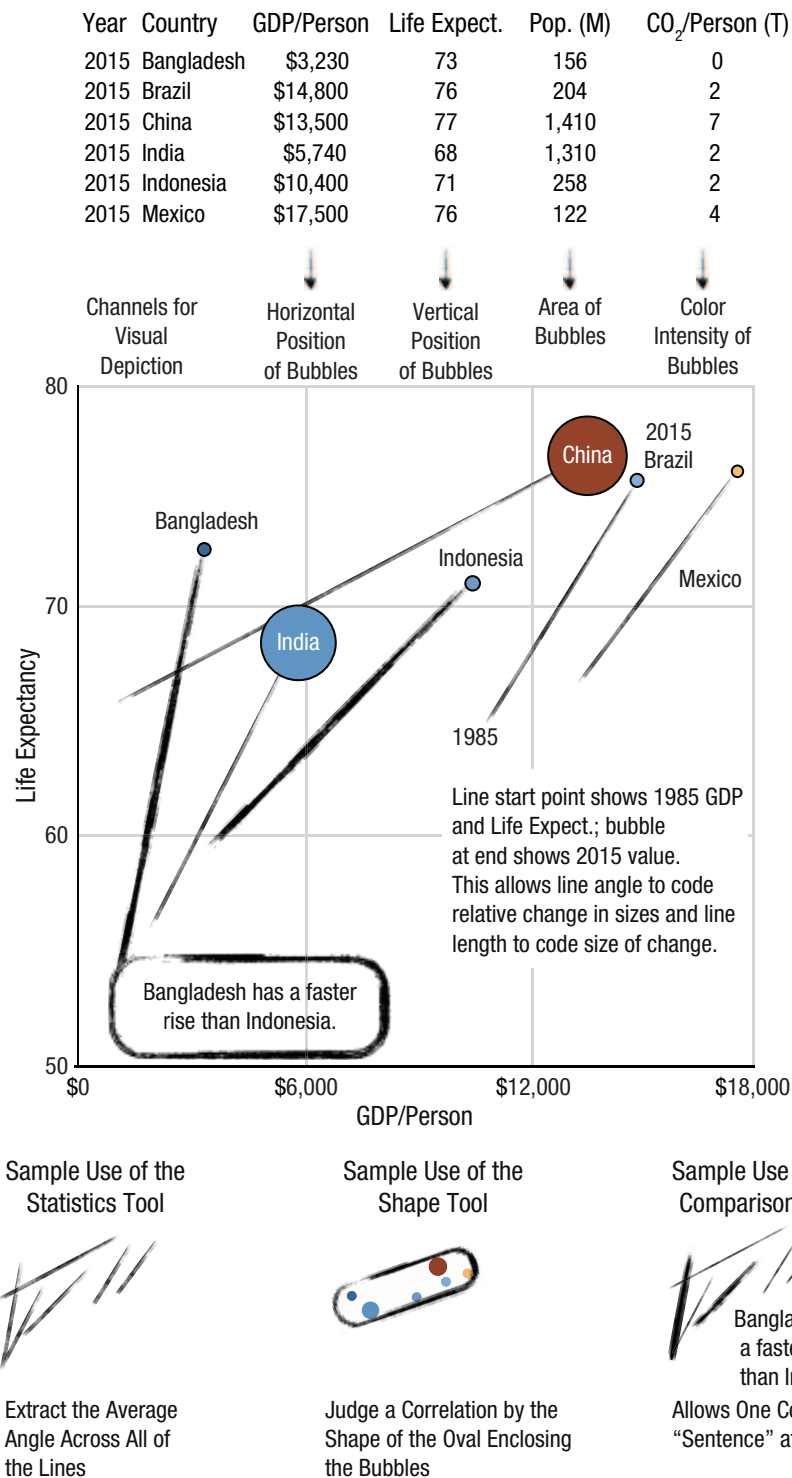


Fig. 1. Illustration of how a data visualization can be designed to capitalize on the visual system's perceptual tools. The table at the top shows gross domestic product (GDP, in U.S. dollars) per person, life expectancy ("Life Expect."), population ("Pop."; in millions), and carbon dioxide (CO₂) emissions per person (in tons) for six countries in 2015. Patterns in the tabular data are difficult to discern, but when each column is mapped to a visual channel, as in the visualization in the middle, the patterns can be extracted much more quickly, even with the additional data about the GDP and life expectancy of each country for 1985, shown via the "comet" lines. The examples at the bottom show how each of the three perceptual tools discussed in this article (the statistics tool, shape tool, and comparison tool) could help a viewer extract patterns from this visualization. The data used for this visualization were retrieved online from Gapminder.org. This visualization was inspired by the work of Hans Rosling (2006). Copyright 2020 by Steven L. Franconeri.

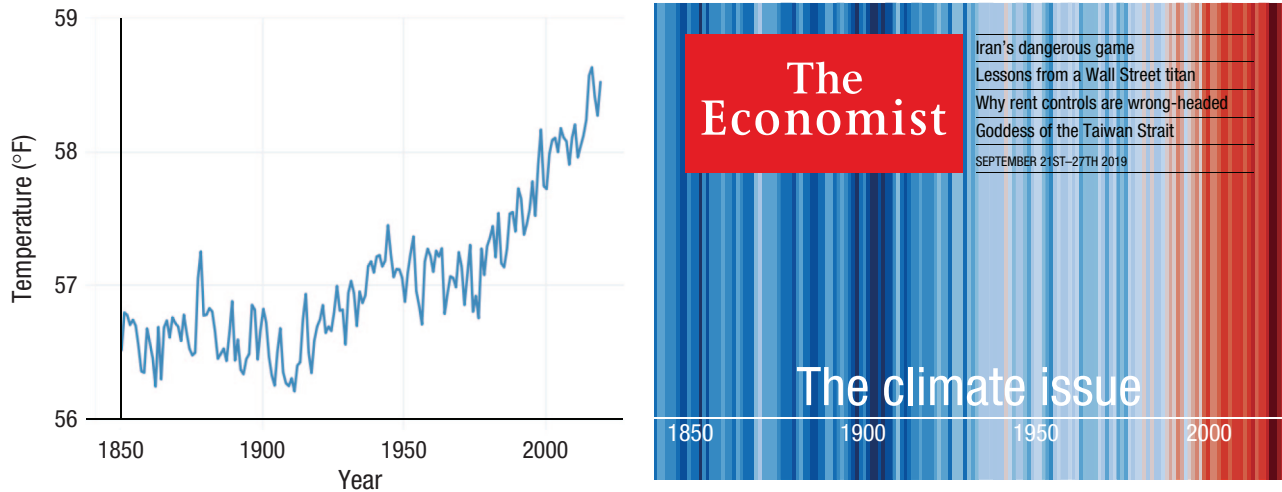


Fig. 2. Example of a visualization design using color intensity to focus the viewer on big-picture statistics. The line graph at the left encodes global climate data precisely. The cover of a September 2019 issue of *The Economist*, shown on the right, used color intensity to depict the same data with far less precision for each value, leading the viewer to focus on the bigger picture.

positions or longer lines are easily processed and understood as reflecting larger values (Tversky, 2001).

In Figure 1, you can quickly extract statistics within any of these channels. Among all of the plotted countries, which has the lowest GDP, coded as horizontal position? Which has the largest population, coded as the bubble's area? How does the rate of change in life expectancy tend to relate to the rate of change in GDP from 1985 to 2015, indicated by the average angle of the tails of the comets (see the leftmost example in the bottom row of Fig. 1)? Within a split second, the visual system can process any of these channels across a two-dimensional page or screen, allowing you to find values or regions in a visualization that differ from others (Wolfe & Horowitz, 2017) or to extract global statistics across those values with impressive precision (Alvarez, 2011). These feats are possible through adaptation of the existing ability of the visual system to quickly extract similar statistics from natural scenes, to help you identify the lowest apple on the tree, the largest slice of pie, or the average angle of ocean waves. These visual statistics, therefore, serve as a proxy for actual statistics, allowing you to extract statistical information from data values that have been mapped to visual channels (for review, see Szafir et al., 2016).

Note that for simple computations of ratios (Jacob et al., 2012), some channels transmit data values more precisely than others (Cleveland & McGill, 1985). For example, in Figure 1, you can easily tell that the ratio of GDP per person between Bangladesh and Indonesia is around 1:3 because horizontal position is precisely extracted on a perceptual level. You are less sure of

the ratio between the CO₂ production of Mexico and China because the color-intensity channel does not allow for precise perception of its underlying values. Although this observation might encourage visualization designers to prefer the more precise channel (position) over the less precise one (color intensity), this assessment of higher precision is based on performance in a single task: judging the ratio between two values at a time. If the task instead requires a viewer to isolate a single value (e.g., find the minimum or maximum), the precision possible with a position channel (e.g., a line graph) may again be better than the precision possible with a color-intensity channel (e.g., a heat map). But for other, more global statistical judgments of the mean or variance of data values, the pattern of relative precision disappears or even flips, so that color intensity beats position (Albers et al., 2014). Figure 2 depicts a real-world example of a design that purposely uses color intensity to focus the viewer on big-picture statistics, instead of a precise reading of individual values.

Despite a large research literature on the processing of visual statistics (see Alvarez, 2011, for review), most previous studies have focused on humans' visual performance within the natural world of scenes, objects, and faces. Recent work in the data-visualization field has begun to adopt these research paradigms to test performance in the artificial world of data visualizations, with the eventual goal of producing a holistic understanding of the precision—and potential biases—of visually recovered statistics (mean, variance, outliers) across actual visualization displays (Szafir et al., 2016).

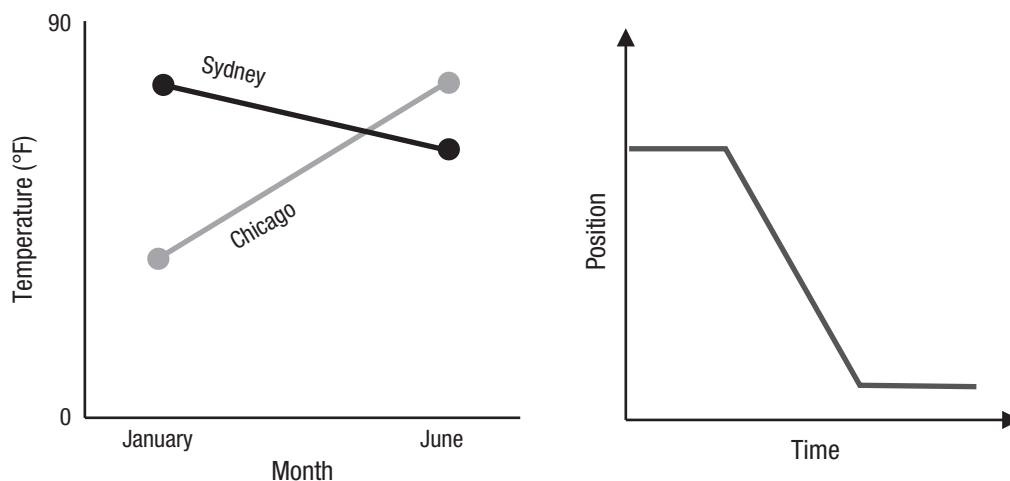


Fig. 3. Examples of the use of shape processing in data visualization. The graph on the left depicts average temperatures for Sydney and Chicago in January and June. Noticing an X shape allows for quick judgment of a critical pattern in the data. The graph on the right, an adaptation of Figure 3 in Kozhevnikov et al. (2002), shows how shape processing can lead viewers astray. Many students viewed the graph as depicting an object that moves horizontally and then down an incline before eventually stopping, but a better interpretation is that it might depict an object that does not move at first and then moves at a constant rate before stopping.

Perceptual Tool 2: Shape Processing

Viewers can also directly map shapes in a visualization to shapes previously seen in past visualizations. Figure 3 (left panel) depicts average temperatures for January and June in two cities. Experienced graph readers will immediately notice the X-shaped pattern in the data, which provides a shortcut for them to understand that the seasons' relative temperatures are "flipped" across the cities. Similarly, when viewing a scatterplot (see the middle example in the bottom row of Fig. 1), people appear to use the shape of the oval that encloses most of the points as a proxy for correlation (Yang et al., 2019).

Shape processing can also lead less experienced graph readers astray. In an experiment by Kozhevnikov et al. (2002), students were asked to give a real-world example that might produce the graphed data shown in the right panel of Figure 3. A correct answer would be "An object does not move for a period of time, then it moves at a constant rate, and then it stops." But shape processing led many students to see the graph as a natural scene of a hill, guiding them toward answers such as "An object rolls along a flat surface, then down a hill, and then it stops."

Figure 4 shows examples in which shape processing can lead to misunderstandings of statistical uncertainty. In the visualizations on the left, statistical estimates of uncertainty are depicted as shapes (a cone of uncertainty for a hurricane's path; error bars for an election forecast). Translating those shapes into an intuitive

sense of uncertainty requires specialized statistical training. Without training, these depictions can lead to serious misunderstandings, such that viewers might interpret the hurricane map, for example, as illustrating that the hurricane will get larger over time, as it moves from its current position, instead of illustrating that predictions of the hurricane's path are more uncertain for the more distant future (Franconeri et al., in press). The visualizations on the right show graphic redesigns that leverage viewers' shape-processing tool and that communicate uncertainty in a more naturalistic way (Franconeri et al., in press).

Perceptual Tool 3: Comparisons

Figure 1 allows you to efficiently browse specific comparisons among individual countries and groups of countries. For example, Bangladesh has a faster rise in life expectancy compared with India, and Mexico produces more GDP per person than India. Note that each of these comparisons is expressed as a sentence. The *direction* of each comparison sentence is psychologically important (Miller & Johnson-Laird, 1976). One is unlikely to say that "the building is to the right of the bike" even though it is logically equivalent to "the bike is to the left of the building." Or compare the excitement of "I met Barack Obama" with the hubris of "Barack Obama met me" (Gleitman et al., 1996). Visual comparisons, including those within data visualizations, appear to have a similar directional structure. Moving

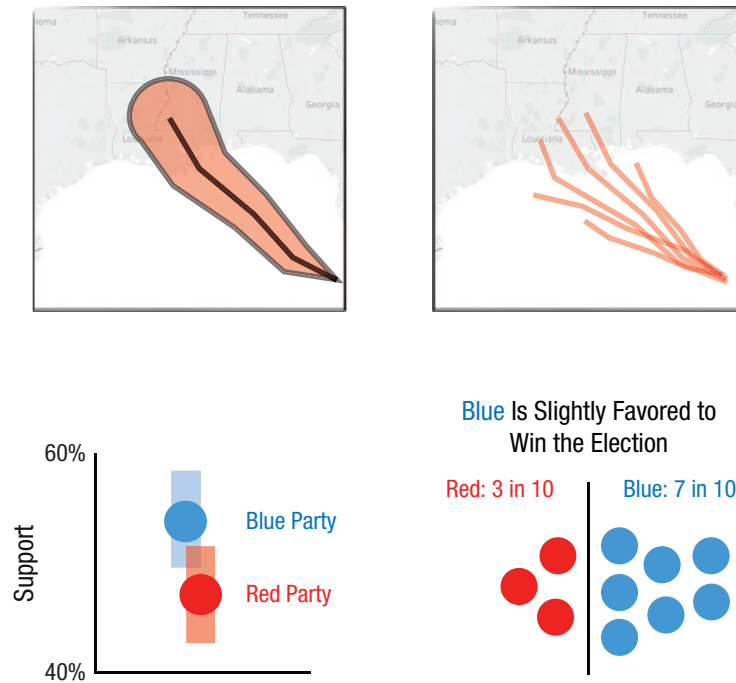


Fig. 4. Examples illustrating how shape processing can lead to misinterpretation of uncertainty and how this tool can be leveraged to facilitate correct interpretation. The top row shows two contrasting ways of presenting the uncertainty of a hurricane's path. The illustration on the left, which uses a shape envelope to reflect statistical variation among different models, could prompt viewers to substantially misinterpret the map as indicating that the hurricane will increase in size as it approaches land, whereas the illustration on the right, which shows a set of example paths from the models, enables observers to extract naturalistic statistics. The illustrations in this row are adaptations of Figure 1 in Liu et al. (2018). The bottom row shows two contrasting ways of presenting the uncertainty in predicting the results of the 2020 U.S. presidential election. The graph on the left uses similar shape envelopes (error bars) to convey uncertainty and is tough to understand without statistical training, whereas the graph on the right, inspired by a style used by *fivethirtyeight.com*, uses a more naturalistic statistical metaphor that leverages people's natural ability to extract statistical summaries from collections of objects.

your eyes or attention from Mexico to India in Figure 1 might tell you that, in 2015, Mexico had a higher GDP per person, whereas the opposite eye movement might tell you that India had a lower GDP per person. When viewers' attention is dragged around a screen by animated cues (Roth & Franconeri, 2012), or roams freely but is monitored with an eye tracker (Michal et al., 2016, Michal & Franconeri, 2017), they are faster to verify the truth of sentences about the information presented (e.g., Mexico has a higher GDP per person than India) if their attention or eyes have traveled between the values in the same directional order expressed in the sentences.

How many unique sentences, and therefore unique comparisons, could you produce from Figure 1? If you start with only two countries, there will be several comparisons (e.g., China has a larger population than

Brazil, China has a lower GDP per person than Brazil; see Michal et al., 2016, for examples of a surprisingly large number of potential comparisons from otherwise simple graphs). The total number of comparisons you could generate from Figure 1 would be enormous, given the number of possible grouping combinations within six countries, four data types, and, in some cases, data for 2 years.

Most models of such relational comparisons predict that you can process only a handful at a time (Hummel & Biederman, 1992), and some predict that the limit is only one at a time (see Franconeri et al., 2012, for review). One recent study validated that this severe capacity limitation on comparisons extends to data visualizations. Participants presented with several pairs of two-bar graphs (bars A and B) were extremely slow at distinguishing pairs with one relationship ($A > B$) from

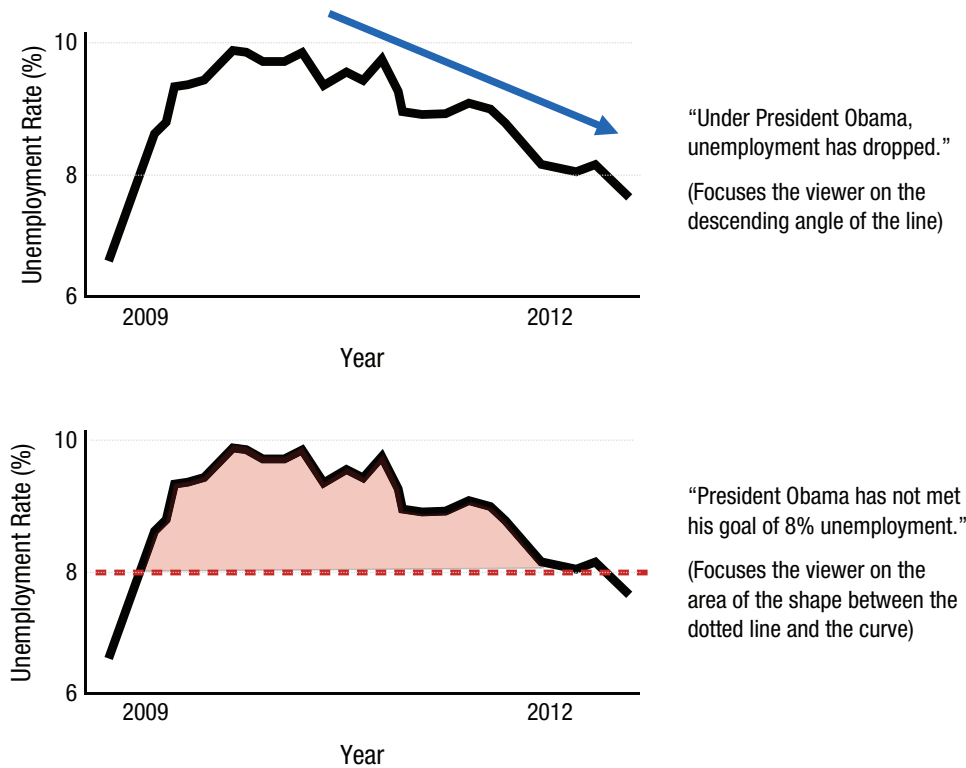


Fig. 5. Illustration of how different ways of presenting data graphically can lead viewers to different conclusions. A viewer can make multiple comparisons within visualized data, and the choice among those comparisons may be as consequential as the choice of a verbal description of the data. A member of U.S. President Obama’s political party (blue) might highlight a positive pattern in the unemployment rate during his administration, whereas a member of the opposite party (red) might highlight a negative pattern. Adapted from “One Report, Diverging Perspectives,” by M. Bostock, S. Carter, A. Cox, and K. Dealy, 2012, October 5, *The New York Times*, <https://archive.nytimes.com/www.nytimes.com/interactive/2012/10/05/business/economy/one-report-diverging-perspectives.html>.

pairs with the opposite relationship ($A < B$) because of the need to compare within each pair. But using the perceptual statistics tool, they could instantly make judgments regarding the same displays, such as identifying the single largest bar (Nothelfer & Franconeri, 2019).

Because comparison is such a relatively slow process, visualization designers often try to help direct viewers to compare the “right” values. One strategy is to use visual grouping cues that encourage the viewer to process those values together (Shah & Hoeffner, 2002). Examples of grouping cues include placing the values to be compared close to one another, making them the same color, or connecting them with a line or contour (Brooks, 2015). Because this practice still leaves a viewer with many potential comparisons to make, guides also encourage designers to highlight critical values with a unique color to draw attention to them, and furthermore to annotate that pattern with a single comparison sentence (as in Fig. 1, which includes a sentence comparing Bangladesh and Indonesia).

This practice is now widely used by professional data communicators in business and journalism. The power of these highlighting and annotation techniques should be clear from Figure 5, which shows how competing designs might draw you toward different comparisons, which can lead you toward drawing importantly different conclusions from the data. With this great power to frame the desired patterns for an audience comes great responsibility for doing so ethically. Researchers have examined the persuasive power of such manipulations, finding that highlighting and annotating a given pattern makes it far more likely to be remembered by a viewer (Ajani et al., 2021). (For a curated list of real-world examples, primarily from journalism, of more and less “truthful” designs that vary across definitions of what “truthful” means, see Cairo, 2016.)

Some creators of visualizations might feel that these additional steps are unnecessary; if they see the pattern in their own visualization, they expect that other people

will see it, too. Unfortunately, designers, authors, teachers, and presenters can be afflicted with the *curse of expertise*, failing to distinguish between what they understand and what an audience understands (Birch & Bloom, 2007). Once people recognize an important comparison in a data set, they focus on that comparison automatically and assume that new viewers will, too. For example, in a recent study, participants were presented with a graph summarizing a simulated election-polling data set and told an interesting story about a data pattern in either the top two lines or the bottom two lines (Fig. 6, left panel). They were then told to forget the story and to annotate the visual pattern that a naive viewer (who did not know the background story) would find most salient. If participants had been able to inhibit their own knowledge, there should have been no difference in the annotated patterns between the two conditions. Instead, participants tended to report that the pattern a naive viewer would find most salient would be the same pattern that was the focus of the story they were told (i.e., about either the top two or the bottom two lines; summaries of multiple participants' drawings are shown in Fig. 6, right panel; Xiong et al., 2020). Clearly, participants' expertise served as a "curse" that affected their predictions of what other people would see in the data.

Conclusion

Merging theory, evidence, and methods from psychology with research and practice in data visualization suggests a set of at least three tools that people rely on to process visualized data: statistical extraction, shape processing, and relational comparison. Viewers can effectively use these tools to explore data—but only when visualizations are designed to effectively leverage these tools, and when viewers are properly trained to know which tool to use when.

Understanding the power, limits, and deployment of these tools can help inspire guidelines for visualization designers so that viewers will grasp the information faster and gain deeper understanding of the critical patterns depicted. Further improvements will be possible as research continues. The statistics tool allows viewers to rapidly extract means or outliers, but the precision and potential biases of those operations is not yet fully understood. When deployed appropriately,

the shape tool can provide a fast and holistic view of the data when shapes are simple, yet researchers currently have almost no understanding of how people process complex shapes, such as the dance of a line graph of a company's stock price. The comparison tool extracts single comparisons at a time, and improved understanding of visual grouping should help guide designs that facilitate those comparisons, in theory, but almost no work has validated that this expectation applies across the complexities of real-world visualizations. These are important—and answerable—questions that are being addressed by a new generation of researchers meeting at the intersection of data visualization and cognitive science.

Recommended Reading

- Franconeri, S., Padilla, L. M., Shah, P. D., Zacks, J. M., & Hullman, J. (in press). (See References). Reviews the broader literature on the psychology of data visualization.
- Shah, P., & Hoeffner, J. (2002). (See References). Reviews how people correctly and incorrectly reason with graphs.
- Szafir, D. A., Haroz, S., Gleicher, M., & Franconeri, S. L. (2016). (See References). Reviews the statistics tool.
- Zacks, J. M., & Franconeri, S. L. (2020). Designing graphs for decision-makers. *Policy Insights From the Behavioral and Brain Sciences*, 7(1), 52–63. Briefly reviews prescriptions for clearly communicating data through visual representations.

Transparency

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ORCID iD

Steven L. Franconeri  <https://orcid.org/0000-0001-5244-9764>

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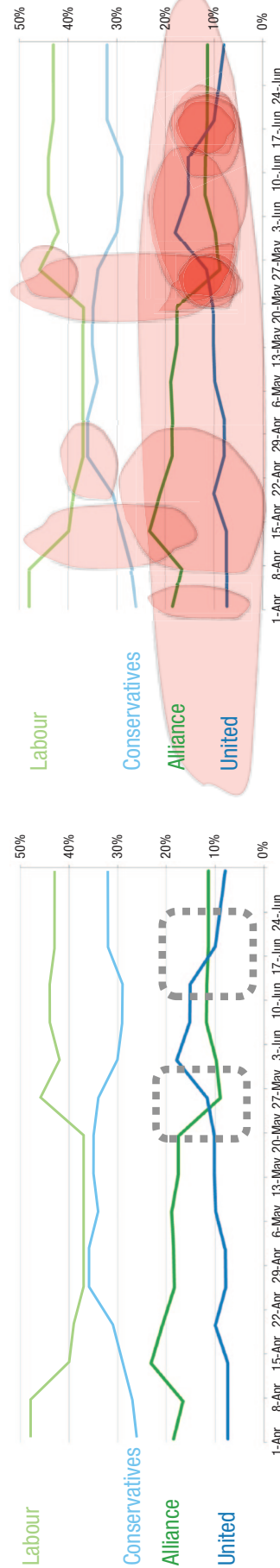
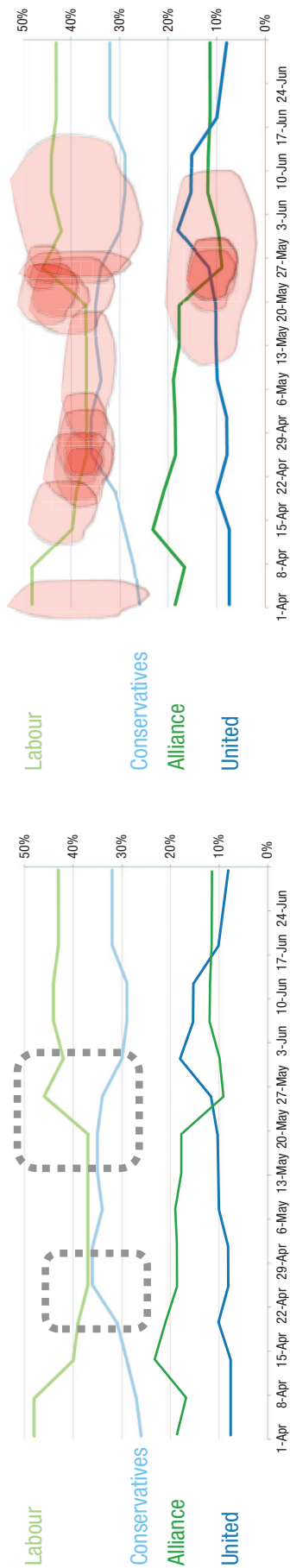


Fig. 6. Example of the curse of expertise in predicting what naive viewers will extract from a data visualization. Participants were shown one of the stimuli on the left and were told a story about the highlighted pattern, in either the top two or the bottom two graphed lines. They were then asked to annotate the graph to show the pattern that a naive viewer would find most salient. In each condition, participants tended to predict that the pattern a naive viewer would find most salient would be the same pattern that had been the focus of the story they had heard, as illustrated by the two combined drawings shown on the right (darker shading indicates that more participants in that condition annotated that region as being salient to other viewers). From "The Curse of Knowledge in Visual Data Communication," by C. Xiong, L. Van Weelden, and S. Franconeri, 2020, *IEEE Transactions on Visualization and Computer Graphics*, 26(10), p. 3053 (<https://doi.org/10.1109/TVCG.2019.2917689>). Copyright 2020 by IEEE. Reprinted with permission.

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